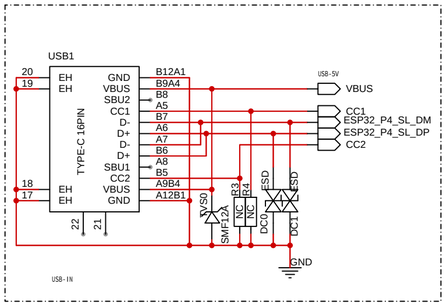
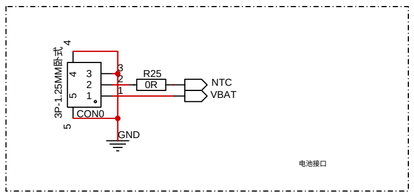
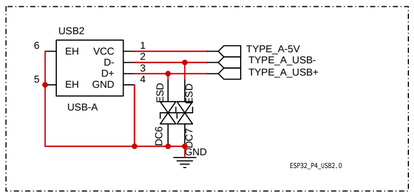
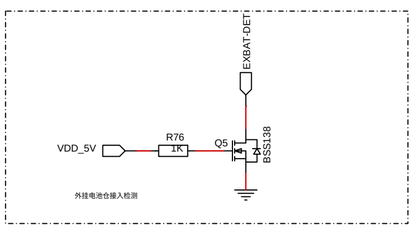
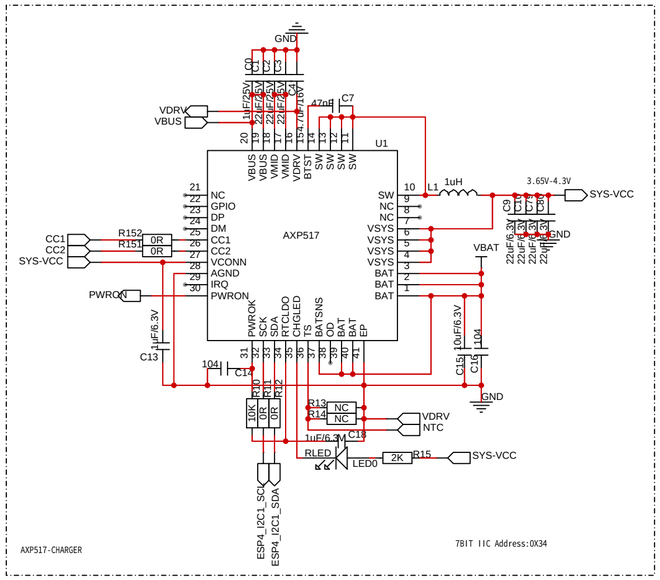
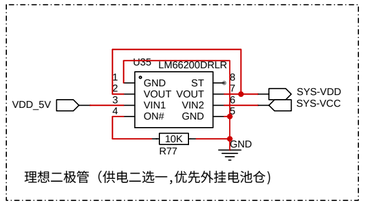


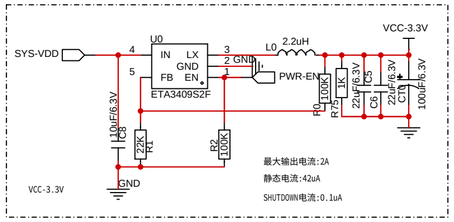


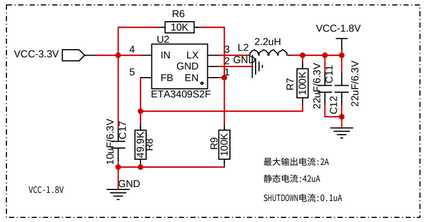


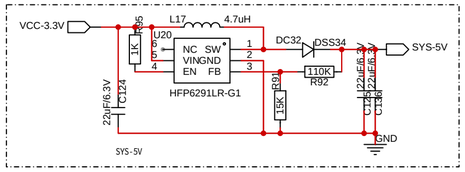


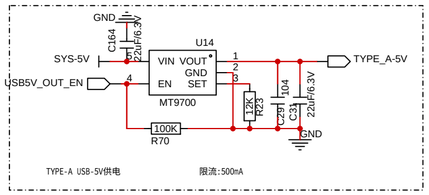


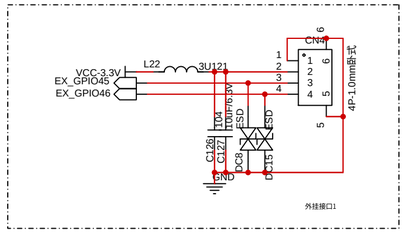


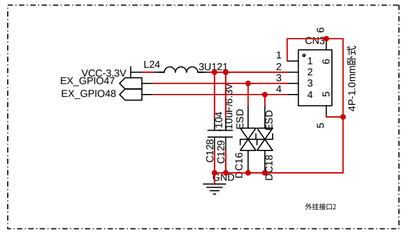


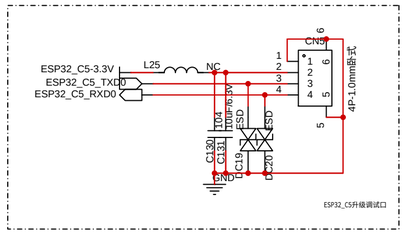


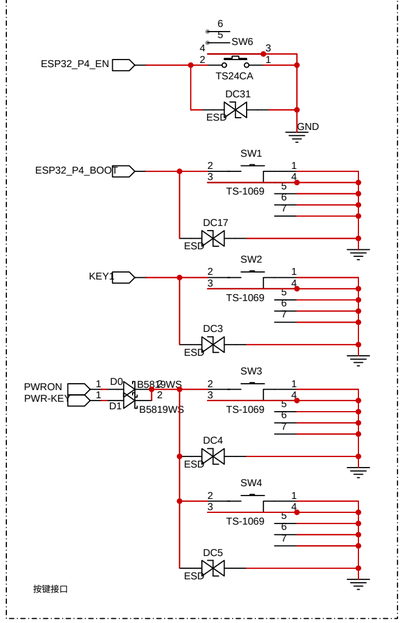


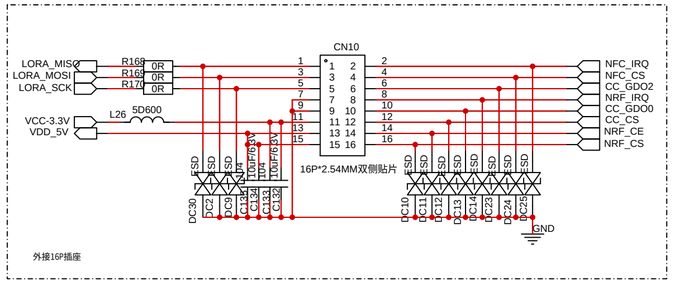






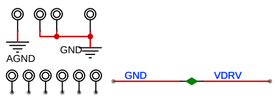






CC_CS:CC1101-CS
CC_GDO0:CC1101-INT
CC_GDO2:CC1101-BUSY
NFC_IRQ:ST25R3916-INT
NFC_CS:ST25R3916-CS
NRF_IRQ:NRF24L01-INT
NRF_CS:NRF24L01-CS
NRF_CE:TXD/RXD-SELECT





ESP32-P4供电

SYS-VDD
10K R30
TMI3181A
L7 2.2uH
ESP32_P4-3.3V
C36 10uF/6.3V
C33 NC
R32 100K
C37 22uF/6.3V
C38 22uF/6.3V
R36 22K
R47 1M
GND
最大输出电流:800mA
静态电流:3uA

40Mhz晶振

X0
ESP32_P4_XN
ESP32_P4_XP
40MHz
C42 12pF
C43 12pF
GND

32.768Khz晶振

C49 12pF
X1 32.768Khz
C51 12pF
ESP32_P4_32K_XIN
ESP32_P4_32K_XOUT
GND

P4状态指示灯

ESP32_P4-3.3V
R45 1K
R100 10K
LED1
BLUE
GND

ESP32-P4NRW32

ESP32_P4_VDD_HP

ESP32_P4-3.3V
U6 TMI3181A
L6 2.2uH
ESP32_P4_VDD_HP
C31 10uF/6.3V
R31 499K
C19 10pF
R33 499K
C23 10uF/6.3V
C24 10uF/6.3V
GND

ESP32_P4_VPST6-1.8V

ESP32_P4-3.3V
SSP7615-18DFR
U31
R143 10K
C161 1uF/6.3V
C40 104
C41 10uF/6.3V
ESP32_P4_VPST6-1.8V
GND

GPIO扩展

I2C Address:0x20

ESP4_I2C1_SDA
ESP4_I2C1_SCL
C39 104
R34 0R
R35 4K7
R38 4K7
R39 NC
VCC-3.3V
U8 XL9535
GPS_RST
GPS_WKUP
ESP32_C5_BOOT
ESP32_C5-EN
SD_VDD-EN
LORA_PCTL
LORA_RST_CTL
BHI260AP_RST_CTL
USB_VCC-EN
TFT_RST_CTRL
SPEAKER-CTL
USB5V_OUT_EN
TP_RST_CTRL
ESP32_P4-LED

待机电流:0.25uA

纽扣电池
ESP32_P4-3.3V
D4 BAT54WS
ESP32_P4_VDD_HP
ESP_LDO_VO4
C45 104
C48 10uF/6.3V
BAT2 NS621FE
RBA
R93 2K
C44 104
C46 104
C47 10uF/6.3V
R40 10K
C50 1uF/6.3V
ESP32_P4_EN
ESP32_P4_32K_XIN
ESP32_P4_XP
ESP32_P4_XN
ESP4_I2C2_SDA
ESP4_I2C2_SCL
ESP32_P4_VPST6-1.8V
TP-INT
BHI260AP_INT
BHI260AP_RST
SD_DET
EX_GPIO48
EX_GPIO47
EX_GPIO46
EX_GPIO45
SD1_CMD
SD1_CLK
SD1_D3
SD1_D2
SD1_D1
SD1_D0
EN_DCDC

U9
GND GPIO0 CHIP_PU VBAT VDDA XTAL_P XTAL_N GPIO54 GPIO53 VDDPST_6 GPIO52 GPIO51 GPIO50 GPIO49 VDD_HP_3 GPIO48 GPIO47 GPIO46 GPIO45 GPIO44 VDDPST_5 GPIO43 GPIO42 GPIO41 GPIO40 GPIO39 EN_DCDC

ESP32_P4_32K_XOUT
LORA_SCK
LORA_MOSI
LORA_MISO
LORA_INT
LORA_BUSY
LORA_NSS
EXBAT-DET
R41 10K
ESP32_P4-3.3V
C52 10uF/6.3V
C53 104
GND
ESP4_I2C1_SDA
ESP4_I2C1_SCL
PWR-KEY
PWR-EN
I2S_MCLK
SD2_D0
SD2_D1
SD2_D2
SD2_D3
SD2_CLK
SD2_CMD
ESP32_P4-3.3V
C65 104
GND
I2S_LRCK
I2S_DSDIN
I2S_ASDOUT
I2S_SCLK
ESP32_P4_VDD_HP
C66 10uF/6.3V
C67 104
GND
GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 VDDPST_1 GPIO9 GPIO10 GPIO11 GPIO12 GPIO13 GPIO14 GPIO15 GPIO16 GPIO17 GPIO18 GPIO19 VDDPST_2 GPIO20 GPIO21 GPIO22 GPIO23 VDD_HP_0

FB_DCDC
VDDPST_DCDC
VDD_HP_2
VDDPST_LDO
VFB/VO4
VFB/VO3
VFB/VO2
VFB/VO1
GPIO38
GPIO37
GPIO36
VDDPST
GPIO35
GPIO34
GPIO33
GPIO32
VDDPST_4
GPIO31
GPIO30
VDDPST
GPIO29
GPIO28
GPIO27
GPIO26
NC
GPIO25
FB_DCDC
ESP32_P4_VDD_HP
ESP32_P4-3.3V
ESP_LDO_VO4
ESP_LDO_VO3
ESP_LDO_VO2
ESP_LDO_VO1
C54 104
C55 104
C56 104
C57 104
C58 104
C59 104
C60 104
C61 104
GND
GPS-TXD
GPS-RXD
GPS_PPS
R42 0R
R43 0R
ESP32_P4_BOOT
R44 10K
R46 10K
C62 104
C63 10uF/6.3V
NRF_CS
NRF_CE
NRF_IRQ
CC_CS
CC_GDO0
CC_GDO2
NFC_CS
NFC_IRQ
KEY1
ESP32_P4_SL_DP
C64 104
GND
ESP32_P4-3.3V
C66 104
R150 0R
ESP32_P4_VDD_HP

最大工作电流:600mA
Lght-sleep工作电流:20uA--3.5mA
Deep-sleep工作电流:5uA--25uA
CPU-EN脚置低工作电流:2uA

R48 4.02K
R49 4.02K
GND
P4_FLASH_CS
P4_FLASH_Q
P4_FLASH_WP
P4_FLASH_HOLD
P4_FLASH_CK
P4_FLASH_D
DSI_D1+
DSI_D1-
DSI_CLK+
DSI_CLK-
DSI_D0+
DSI_D0-
CSI_D0-
CSI_D0+
CSI_CLK+
CSI_CLK-
CSI_D1-
CSI_D1+
ESP_P4_USB-
ESP_P4_USB+
USB_VCCA
ESP32_P4_SL_DM
C68 1uF/6.3V
C69 104
C70 1uF/6.3V
C71 104
GND
ESP_LDO_VO1
ESP_LDO_VO3

ESP32-P4-FLASH/128M

U10 W25Q128JV
10K R54
P4_FLASH_CS
P4_FLASH_Q
P4_FLASH_WP
CS#
SO(IO1)
WP#(IO2)
VSS
EP
VCC
HOLD#/RESET#(IO3)
SCLK
SI(IO0)
ESP_LDO_VO1
P4_FLASH_HOLD
P4_FLASH_CK
P4_FLASH_D
C74 104
GND

USB-PHY供电

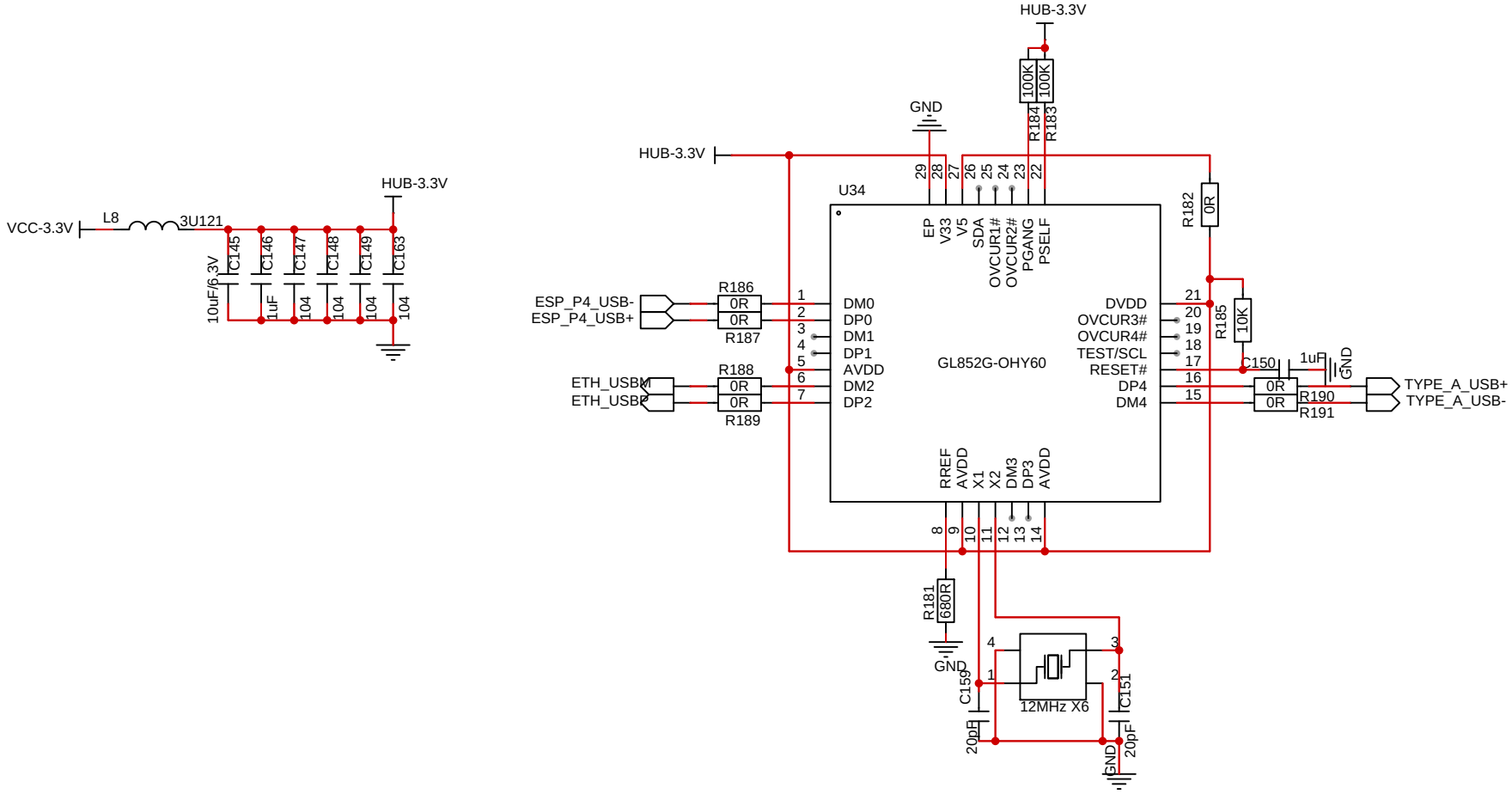
U11 MT9700-N
ESP32_P4-3.3V
C100 10uF/6.3V
R124 10K
GND
USB_VCC-EN
NC R94
NC R58
R57 4.7K
C75 1uF/6.3V
C76 104
C143 103
GND
USB_VCCA

SD卡供电

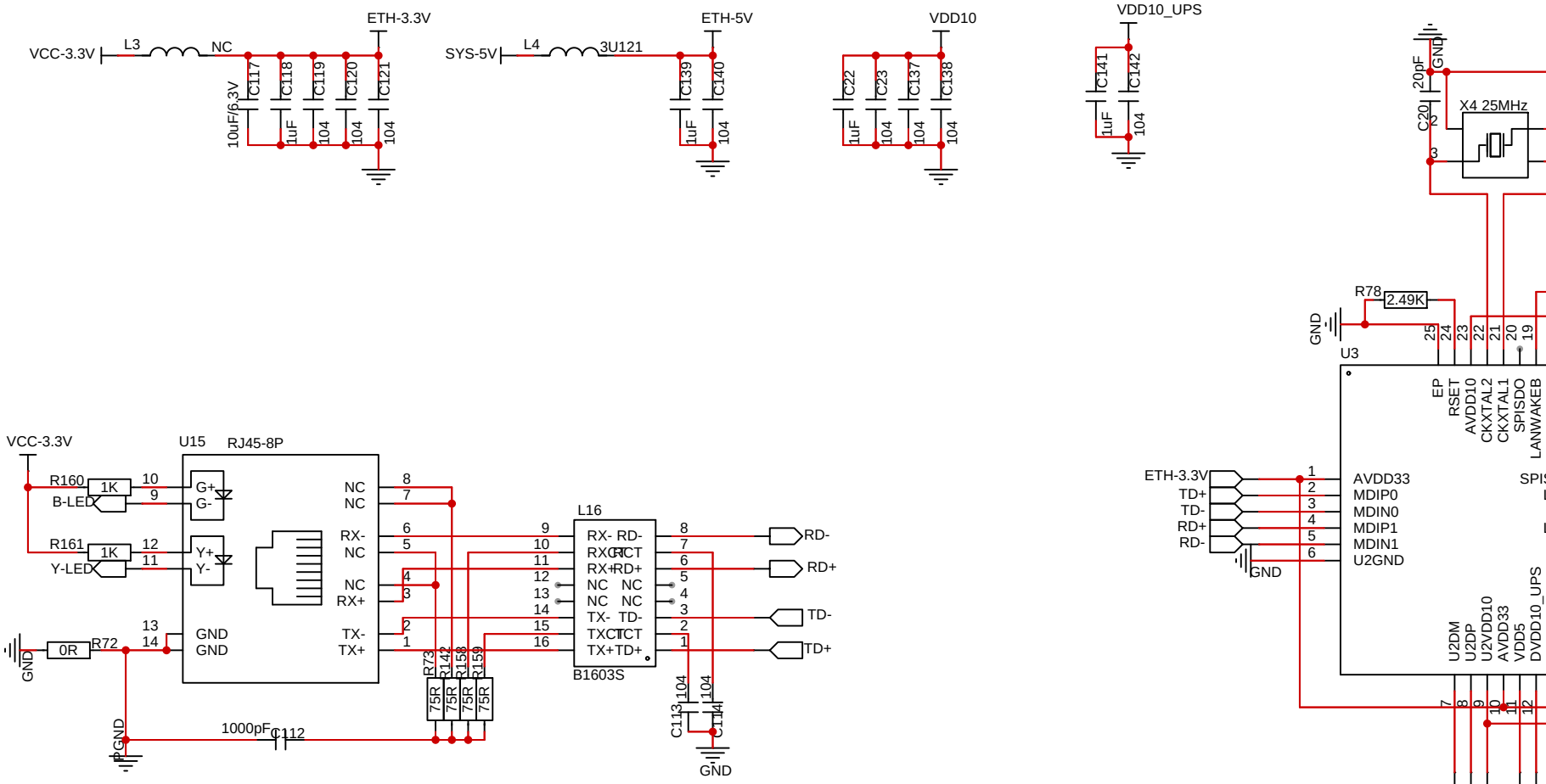
U12 MT9700-N
ESP32_P4-3.3V
SD_VDD-EN
C166 10uF/6.3V
GND
100K R59
R97 4.7K
C77 1uF/6.3V
C78 104
GND
SD_VDD

ESP32_P4-TF

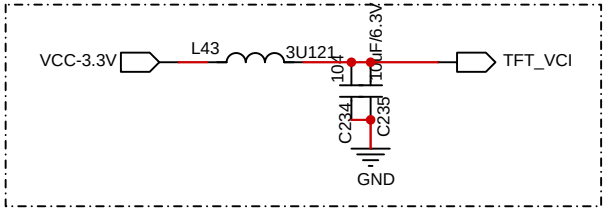
ESP_LDO_VO4
R53 51K
R52 51K
R51 51K
R50 51K
R49 51K
TF1 TF-115-BCP9
SD1_D2
SD1_D3
SD1_CMD
SD_VDD
SD1_CLK
R26 10R
SD1_D0
SD1_D1
SD_DET
DATA2
CD/DATA3
CMD
VDD
CLK
VSS
DAT0
DAT1
CD
GND
GND
GND
GND
DC29 ESD
DC28 ESD
DC27 ESD
DC26 ESD
DC21 ESD
DC22 ESD
C83 NC
GND

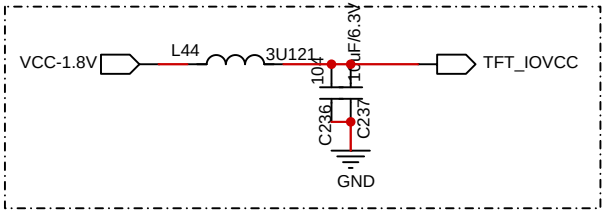


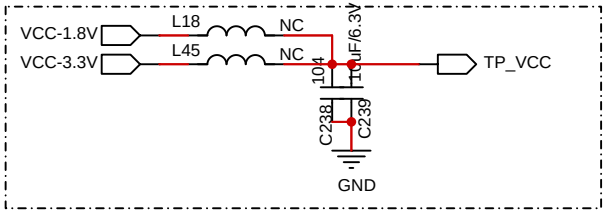
USB-HUB

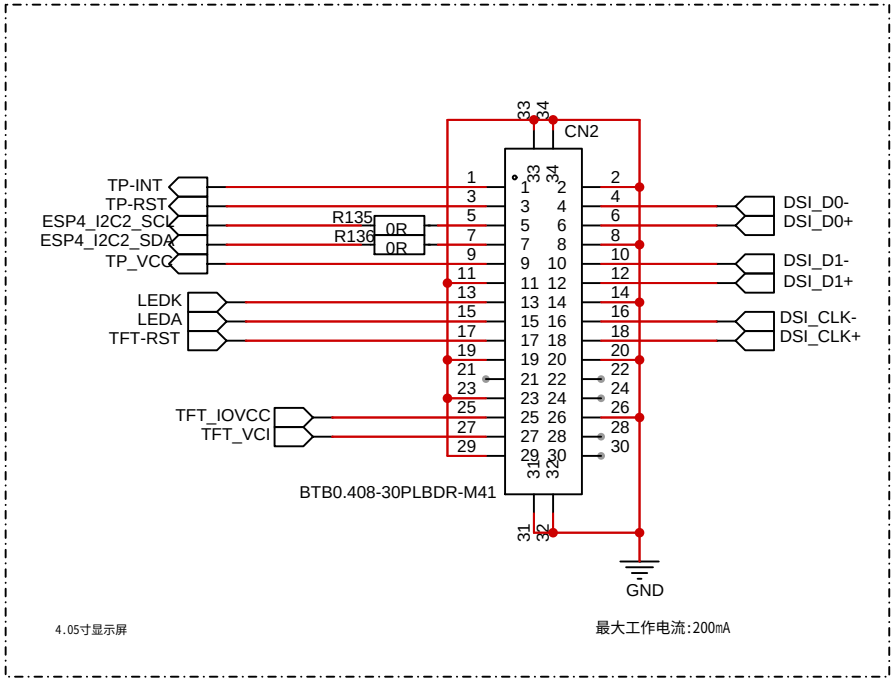


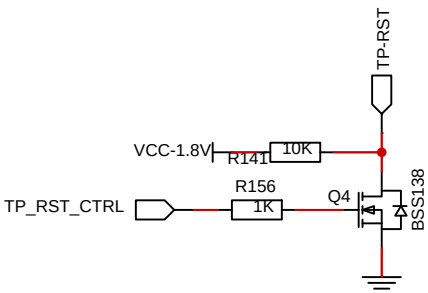
ETH(10/100MHz以太网)

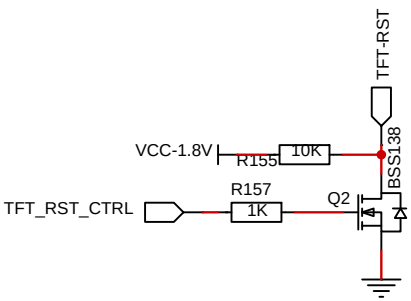

VCC-3.3V
L43
3U121
104
10uF/6.3V
TFT_VCI
C234
C235
GND

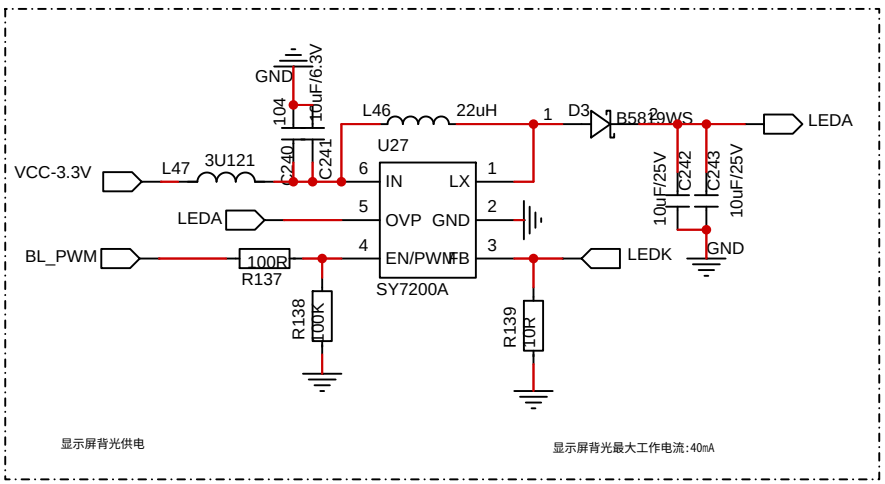

VCC-1.8V
L44
3U121
104
10uF/6.3V
TFT_IOVCC
C236
C237
GND

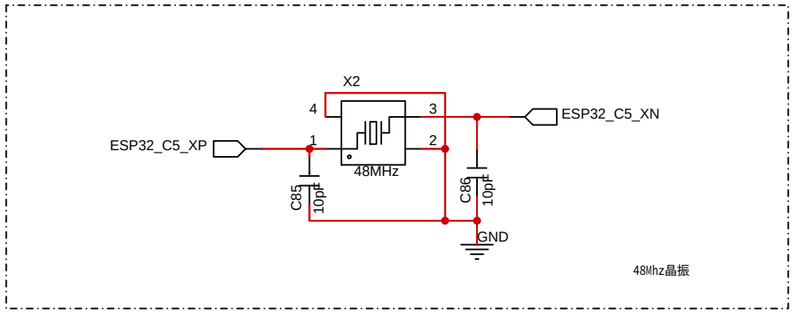

VCC-1.8V
VCC-3.3V
L18
NC
L45
NC
104
10uF/6.3V
TP_VCC
C238
C239
GND

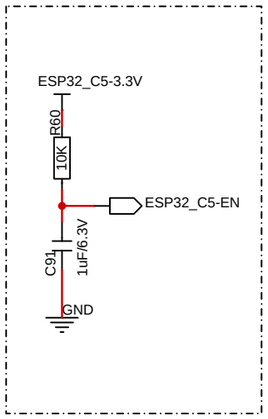

CN2
TP-INT
TP-RST
ESP4_I2C2_SCL
ESP4_I2C2_SDA
TP_VCC
R135
R136
0R
0R
LEDK
LEDA
TFT-RST
TFT_IOVCC
TFT_VCI
DSI_D0-
DSI_D0+
DSI_D1-
DSI_D1+
DSI_CLK-
DSI_CLK+
BTB0.408-30PLBDR-M41
GND
4.05寸显示屏
最大工作电流:200mA

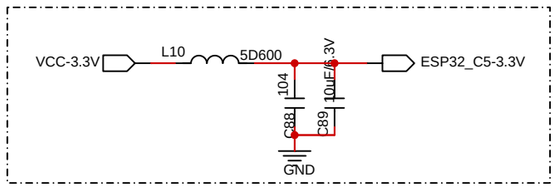

TP-RST
VCC-1.8V
R141
10K
TP_RST_CTRL
R156
1K
Q4
BSS138

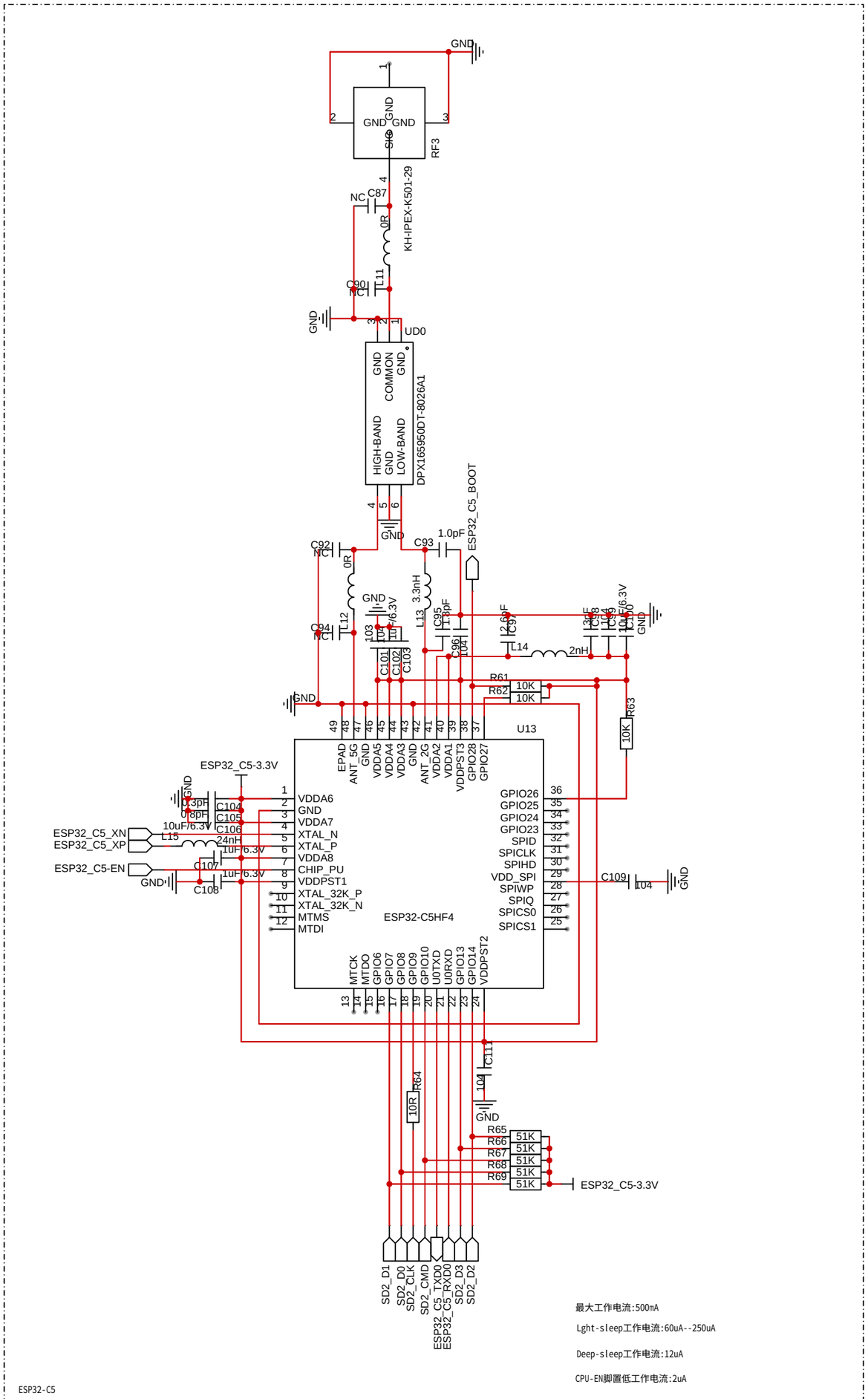

TFT-RST
VCC-1.8V
R155
10K
BSS138
TFT_RST_CTRL
R157
1K
Q2

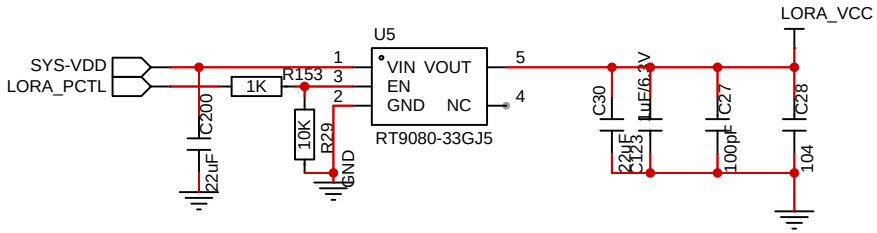

GND
104
10uF/6.3V
L46
22uH
D3
B5819WS
LEDA
U27
VCC-3.3V
L47
3U121
C240
C241
IN
LX
OVP
GND
LEDA
EN/PWM
FB
BL_PWM
100R
R137
LEDK
SY7200A
10uF/25V
C242
C243
10uF/25V
GND
R138
100K
R139
10R
显示屏背光供电
显示屏背光最大工作电流:40mA


X2
4
3
1
2
48MHz
ESP32_C5_XP
ESP32_C5_XN
C85
10pF
C86
10pF
GND
48Mhz晶振

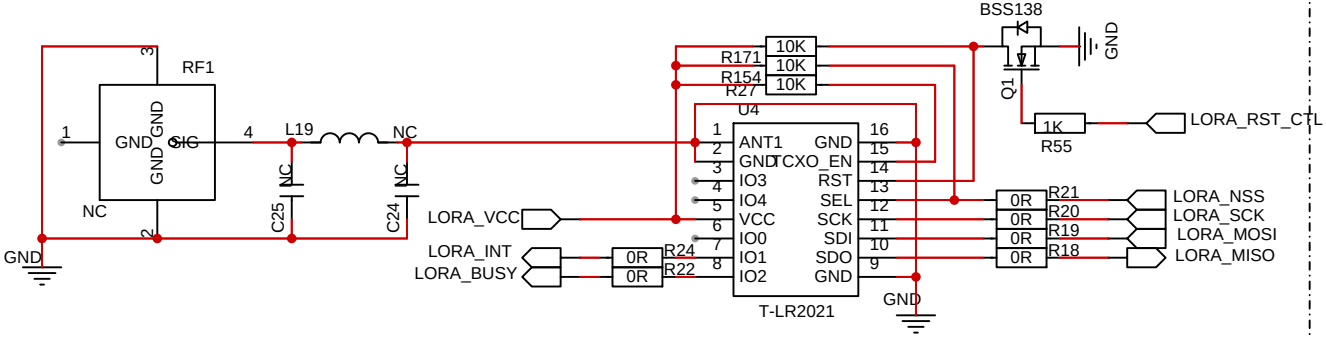

ESP32_C5-3.3V
R60
10K
ESP32_C5-EN
C91
1uF/6.3V
GND


VCC-3.3V
L10
5D600
ESP32_C5-3.3V
104
10uF/6.3V
C88
C89
GND


GND
1
2
GND
GND
3
GND
RF3
NC
C87
4
0R
KH-IPEX-K501-29
C90
NC
L11
GND
3
2
1
UD0
GND
COMMON
GND
HIGH-BAND
GND
LOW-BAND
DPX165950DT-8026A1
4
5
6
GND
ESP32_C5_BOOT
C92
NC
0R
C93
1.0pF
GND
C94
NC
L12
103
104
10uF/6.3V
L13
3.3nH
C95
1.8pF
C96
104
C97
1.6pF
L14
2nH
C98
3pF
C99
104
C100
10uF/6.3V
GND
C101
C102
C103
R61
R62
10K
10K
GND
R63
10K
49
48
47
46
45
44
43
42
41
40
39
38
37
U13
EPAD
ANT_5G
GND
VDDA5
VDDA4
VDDA3
GND
ANT_2G
VDDA2
VDDA1
VDDPST3
GPIO28
GPIO27
ESP32_C5-3.3V
GND
0.3pF
C104
0.8pF
C105
10uF/6.3V
C106
ESP32_C5_XN
ESP32_C5_XP
L15
24nH
10uF/6.3V
C107
ESP32_C5-EN
GND
10uF/6.3V
C108
1
2
3
4
5
6
7
8
9
10
11
12
VDDA6
GND
VDDA7
XTAL_N
XTAL_P
VDDA8
CHIP_PU
VDDPST1
XTAL_32K_P
XTAL_32K_N
MTMS
MTDI
ESP32-C5HF4
GPIO26
GPIO25
GPIO24
GPIO23
SPID
SPICLK
SPIHD
VDD_SPI
SPIWP
SPIQ
SPICS0
SPICS1
36
35
34
33
32
31
30
29
28
27
26
25
C109
104
GND
13
14
15
16
17
18
19
20
21
22
23
24
MTCK
MTDO
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
U0TXD
U0RXD
GPIO13
GPIO14
VDDPST2
C111
104
GND
R64
10R
R65
R66
R67
R68
R69
51K
51K
51K
51K
51K
ESP32_C5-3.3V
SD2_D1
SD2_D0
SD2_CLK
SD2_CMD
ESP32_C5_TXD0
ESP32_C5_RXD0
SD2_D3
SD2_D2
最大工作电流:500mA
Lght-sleep工作电流:60uA--250uA
Deep-sleep工作电流:12uA
CPU-EN脚置低工作电流:2uA
ESP32-C5

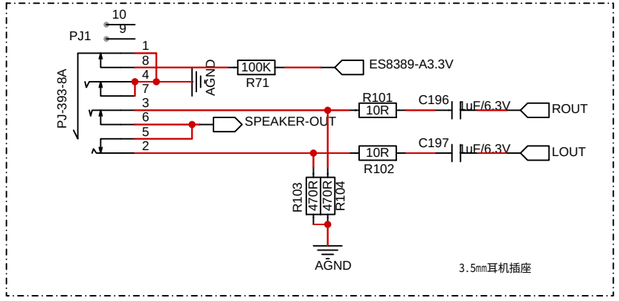


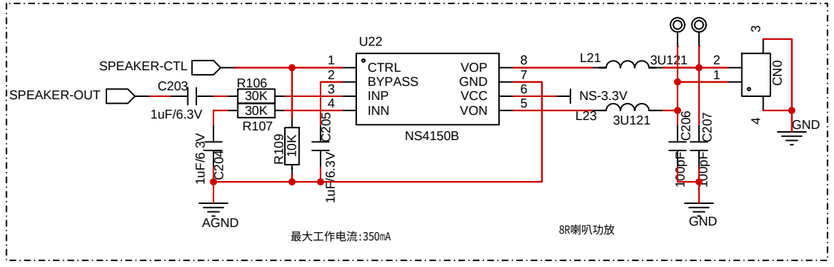
LORA-VDD供电

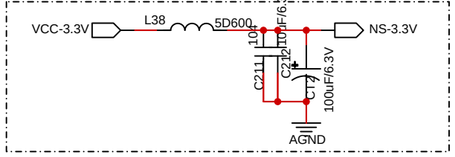


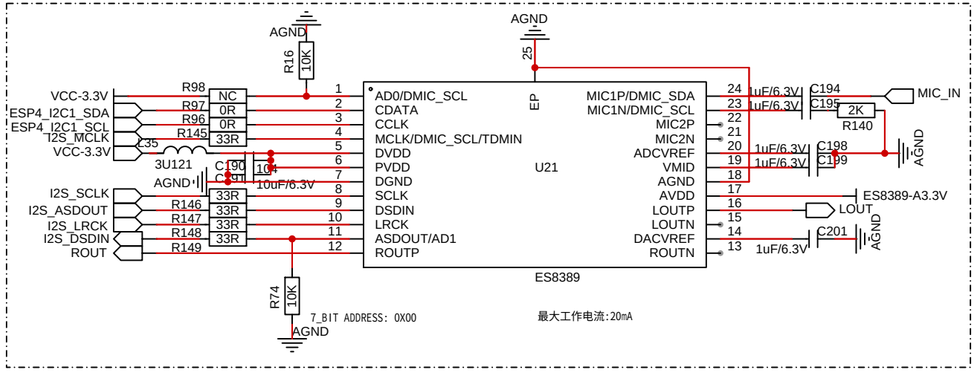
LORA

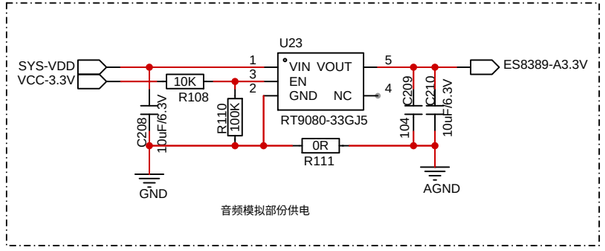
$$$23374 GND

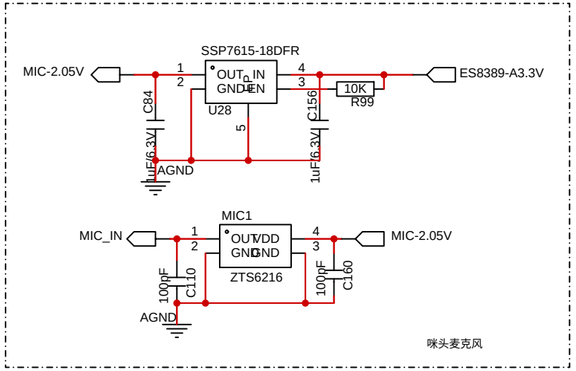

PJ1
PJ-393-8A
AGND
100K
R71
ES8389-A3.3V
SPEAKER-OUT
R101
10R
C196
uF/6.3V
ROUT
R102
C197
LOUT
R103
470R
R104
3.5mm耳机插座

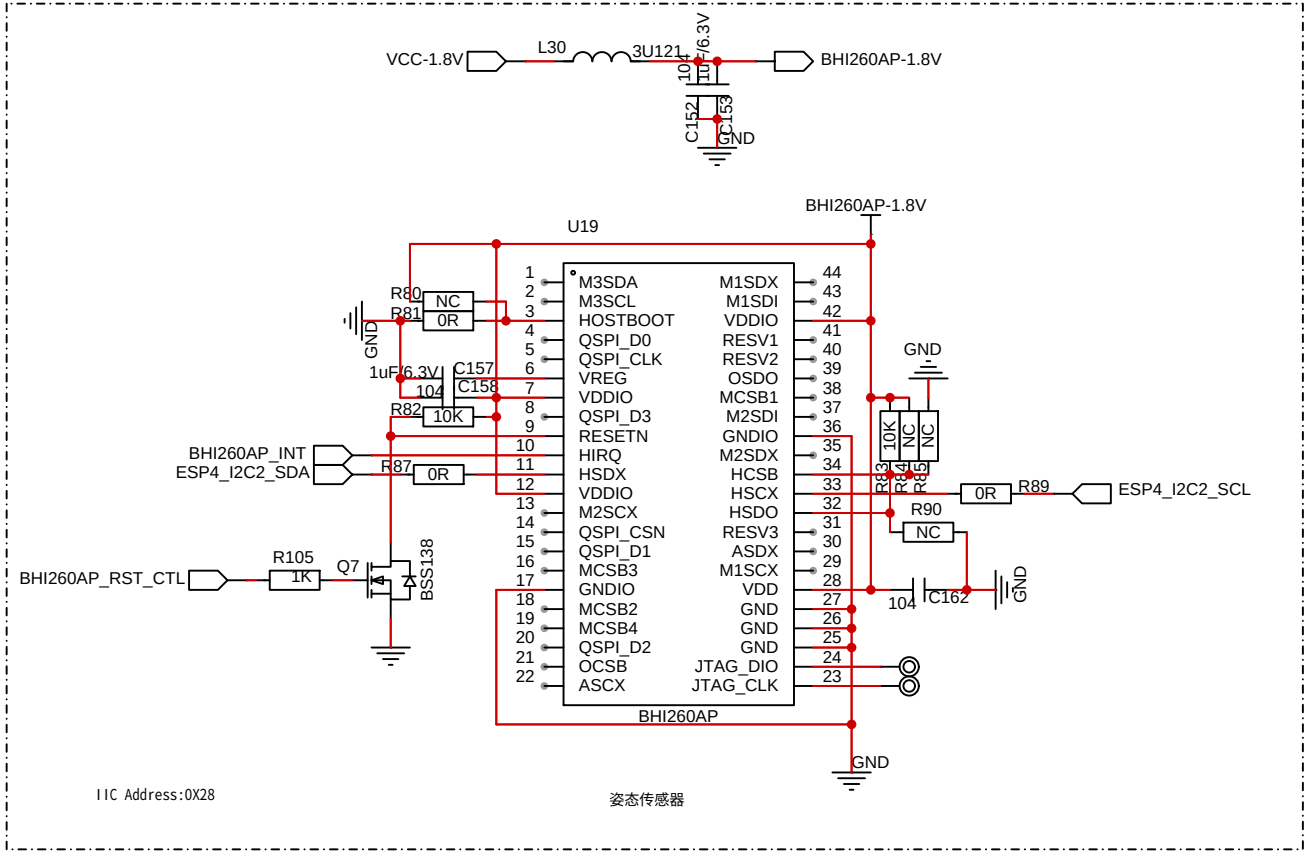

U22
SPEAKER-CTL
SPEAKER-OUT
C203
1uF/6.3V
R106
30K
R107
CTRL
BYPASS
INP
INN
VOP
GND
VCC
VON
NS4150B
R109
10K
C204
C205
L21
3U121
NS-3.3V
L23
C206
C207
CN0
GND
最大工作电流:350mA
8R喇叭功放

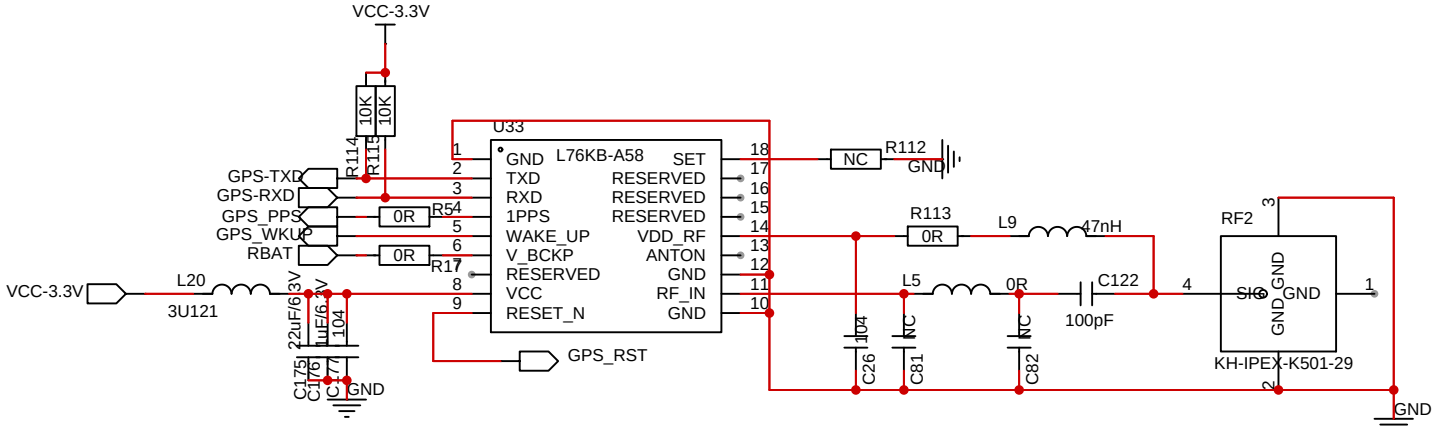

VCC-3.3V
L38
5D600
NS-3.3V
C211
C212
C213
100uF/6.3V
AGND

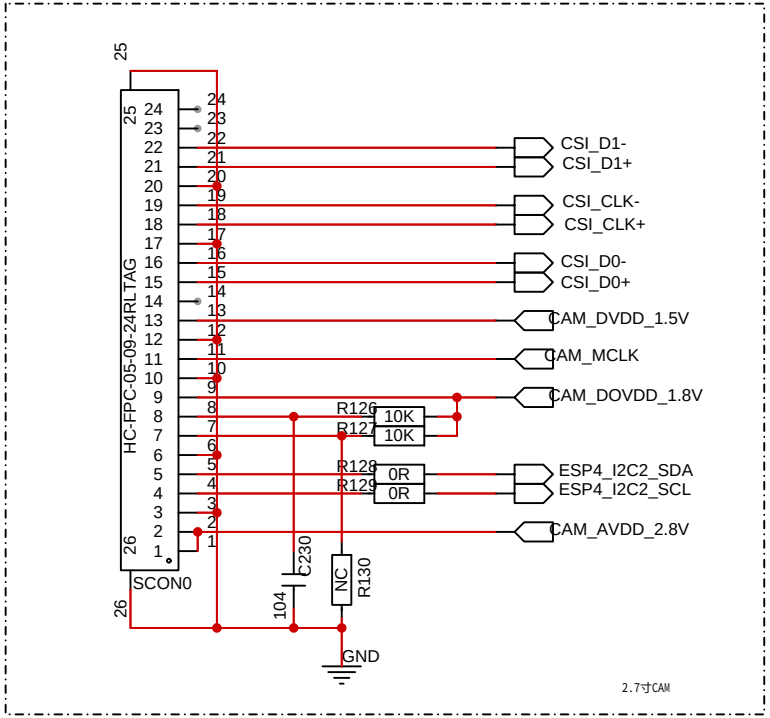

AGND
R16
10K
VCC-3.3V
ESP4_I2C1_SDA
ESP4_I2C1_SCL
I2S_MCLK
R98
R97
R96
R145
NC
0R
0R
33R
L35
3U121
C190
104
C191
10uF/6.3V
I2S_SCLK
I2S_ASDOUT
I2S_LRCK
I2S_DSDIN
ROUT
R146
R147
R148
R149
33R
AD0/DMIC_SCL
CDATA
CCLK
MCLK/DMIC_SCL/TDMIN
DVDD
PVDD
DGND
SCLK
DSDIN
LRCK
ASDOUT/AD1
ROUTP
EP
U21
MIC1P/DMIC_SDA
MIC1N/DMIC_SCL
MIC2P
MIC2N
ADCVREF
VMID
AGND
AVDD
LOUTP
LOUTN
DACVREF
ROUTN
1uF/6.3V
C194
C195
2K
R140
MIC_IN
C198
C199
ES8389-A3.3V
LOUT
C201
ES8389
R74
10K
7_BIT ADDRESS: 0X00
最大工作电流:20mA

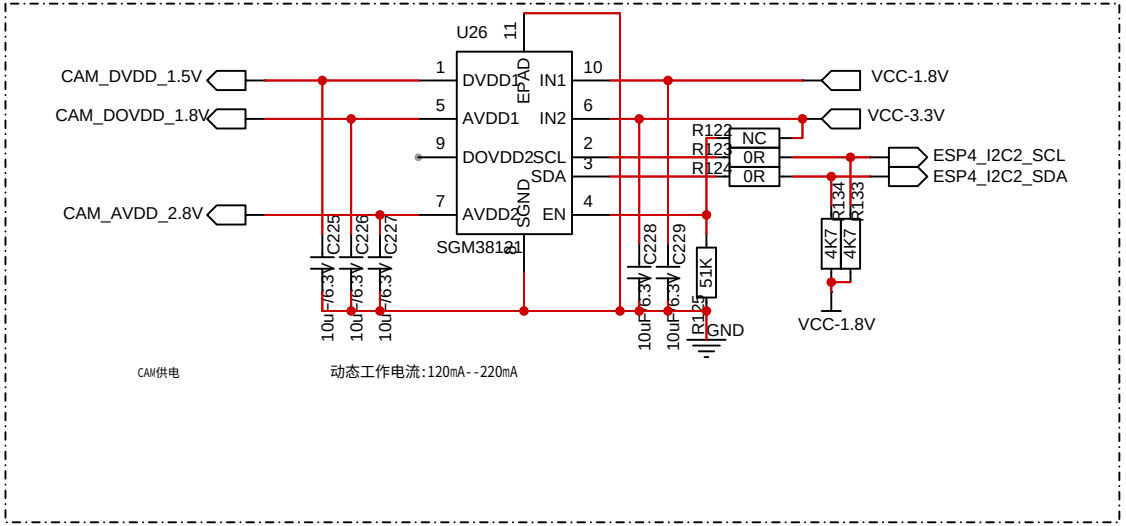

U23
SYS-VDD
VCC-3.3V
10K
R108
VIN
EN
GND
VOUT
NC
RT9080-33GJ5
ES8389-A3.3V
C208
10uF/6.3V
R110
100K
0R
R111
104
C209
C210
10uF/6.3V
GND
AGND
音频模拟部份供电

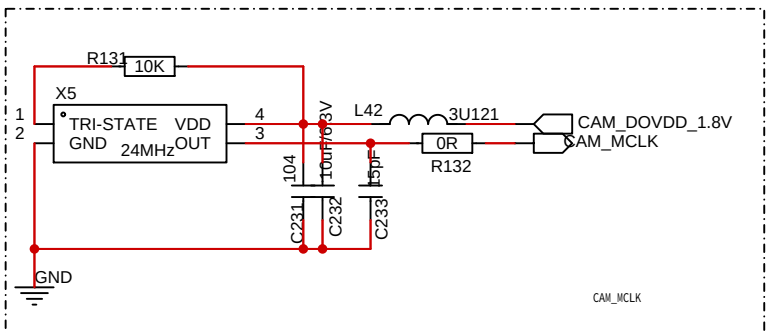

SSP7615-18DFR
MIC-2.05V
OUT
GND
IN
EN
U28
C84
uF/6.3V
AGND
10K
R99
ES8389-A3.3V
C156
1uF/6.3V
MIC1
MIC_IN
VDD
GND
ZTS6216
MIC-2.05V
C110
100nF
C160
咪头麦克风

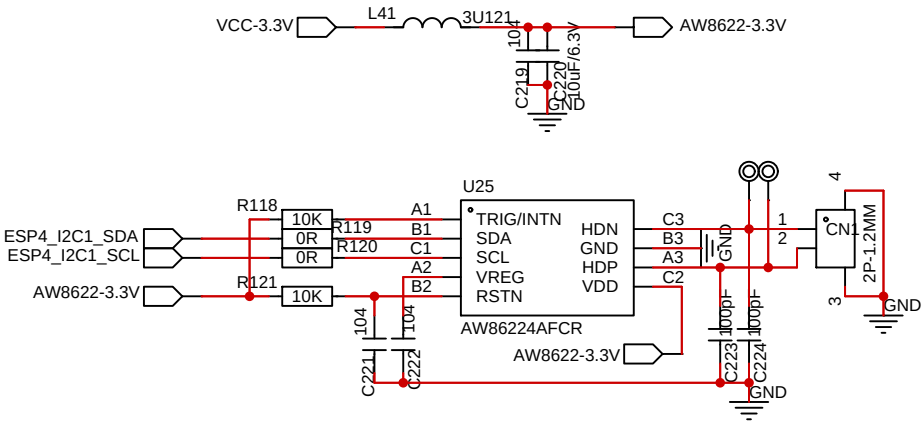

VCC-1.8V
L30
3U121
104
10uF/6.3V
BHI260AP-1.8V
C152
C153
GND
BHI260AP-1.8V
U19
R80
R81
NC
0R
GND
1uF/6.3V
C157
104
C158
R82
10K
BHI260AP_INT
ESP4_I2C2_SDA
R87
0R
BHI260AP_RST_CTL
R105
1K
Q7
BSS138
M3SDA
M3SCL
HOSTBOOT
QSPI_D0
QSPI_CLK
VREG
VDDIO
QSPI_D3
RESETN
HIRQ
HSDX
VDDIO
M2SCX
QSPI_CSN
QSPI_D1
MCSB3
GNDIO
MCSB2
MCSB4
QSPI_D2
OCSB
ASCX
M1SDX
M1SDI
VDDIO
RESV1
RESV2
OSDO
MCSB1
M2SDI
GNDIO
M2SDX
HCSB
HSCX
HSDO
RESV3
ASDX
M1SCX
VDD
GND
GND
GND
JTAG_DIO
JTAG_CLK
BHI260AP
GND
10K
NC
NC
R93
R94
R95
0R
R89
ESP4_I2C2_SCL
R90
NC
104
C162
GND
GND
IIC Address:0X28
姿态传感器
VCC-3.3V
L31
3U121
104
10uF/6.3V
QM-3.3V
C154
C155
GND
U30
QMC_SCL
GND
A1
A2
0R
R88
VSS
SCL
SDA
VDD
QMC6309
B2
B1
R86
0R
QM-3.3V
QMC_SDA
3D磁传感器
U16
10K
R28
QMC_SDA
VCC-1.8V
ESP4_I2C2_SDA
B2
GND
VCCA
A2
B1
VCCB
OE
A1
RS0102YVS8
QMC_SCL
VCC-3.3V
ESP4_I2C2_SCL
104
C115
104
C116
GND


VCC-3.3V
10K
10K
R114
R115
U33
L76KB-A58
GND
TXD
RXD
1PPS
WAKE_UP
V_BCKP
RESERVED
VCC
RESET_N
SET
RESERVED
RESERVED
VDD_RF
ANTON
GND
RF_IN
GND
GPS-TXD
GPS-RXD
GPS_PPS
GPS_WKUP
RBAT
0R
R54
0R
R17
GPS_RST
NC
R112
GND
R113
0R
L9
47nH
L5
0R
C122
100pF
104
C26
NC
C81
NC
C82
RF2
SIG
GND
GND
GND
KH-IPEX-K501-29
GND
VCC-3.3V
L20
3U121
22uF/6.3V
1uF/6.3V
104
C175
C176
C177
GND
模块最大工作电流:100mA
RTC工作电流:8uA


HC-FPC-05-09-24RLTAG
SCON0
CSI_D1-
CSI_D1+
CSI_CLK-
CSI_CLK+
CSI_D0-
CSI_D0+
CAM_DVDD_1.5V
CAM_MCLK
CAM_DOVDD_1.8V
CAM_AVDD_2.8V
ESP4_I2C2_SDA
ESP4_I2C2_SCL
R126 10K
R127 10K
R128 0R
R129 0R
C230 104
NC R130
GND
2.7寸CAM


U26
SGM38121
EPAD
SGND
DVDD1
AVDD1
DOVDD2
AVDD2
IN1
IN2
SCL
SDA
EN
CAM_DVDD_1.5V
CAM_DOVDD_1.8V
CAM_AVDD_2.8V
VCC-1.8V
VCC-3.3V
ESP4_I2C2_SCL
ESP4_I2C2_SDA
R122 NC
R123 0R
R124 0R
R134 4K7
R133 4K7
C225 10u/6.3V
C226 10u/6.3V
C227 10u/6.3V
C228 10uF/6.3V
C229 10uF/6.3V
R125 51K
GND
CAM供电
动态工作电流:120mA--220mA


R131 10K
X5
TRI-STATE
GND
VDD
24MHz
OUT
L42 3U121
0R R132
104 C231
10uF/6.3V C232
15pF C233
CAM_DOVDD_1.8V
CAM_MCLK
GND
CAM_MCLK



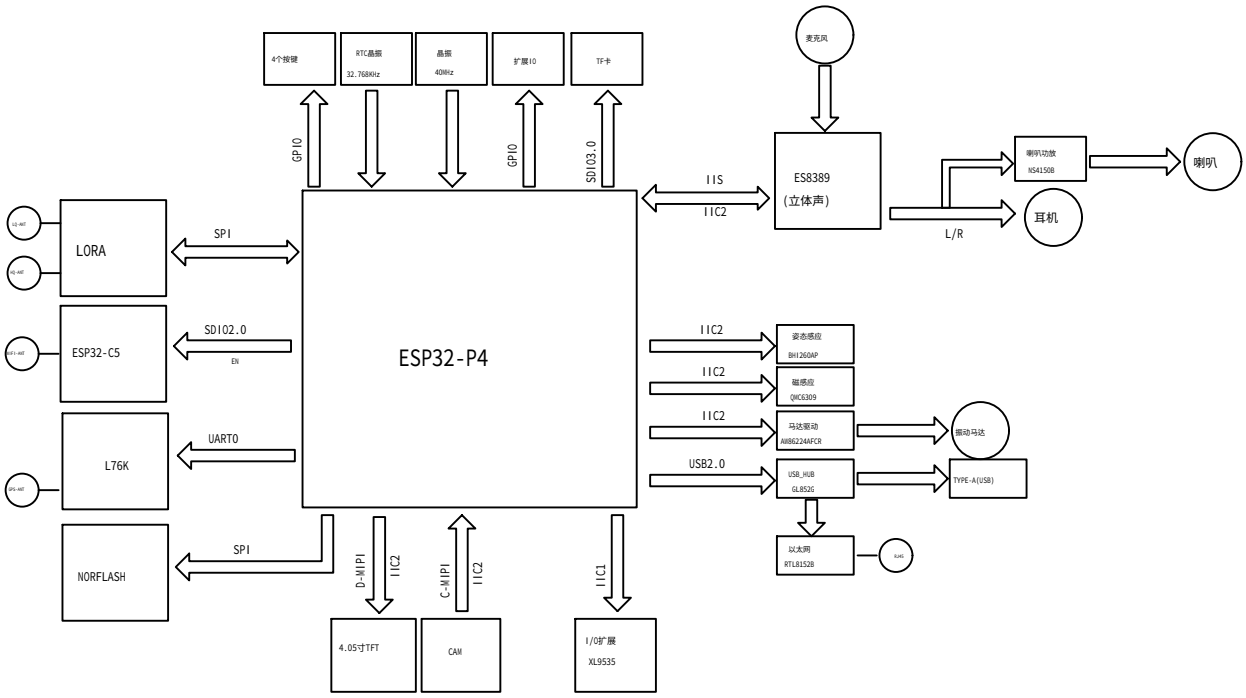
匹配马达额定工作电流:130mA

Standby工作电流:3uA

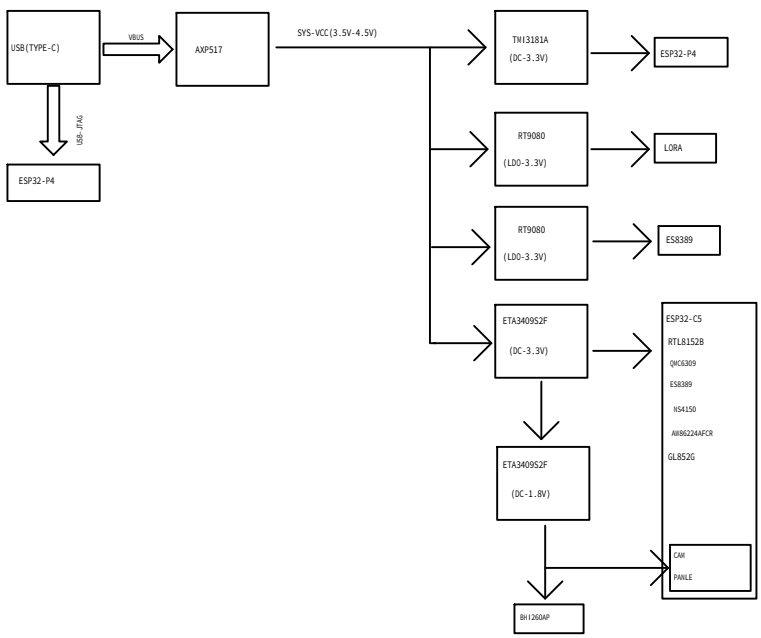
Shutdown工作电流:<1uA

Quiescent工作电流:1.5mA

马达驱动



BLOCK DIAGRAM



POWER DIAGRAM